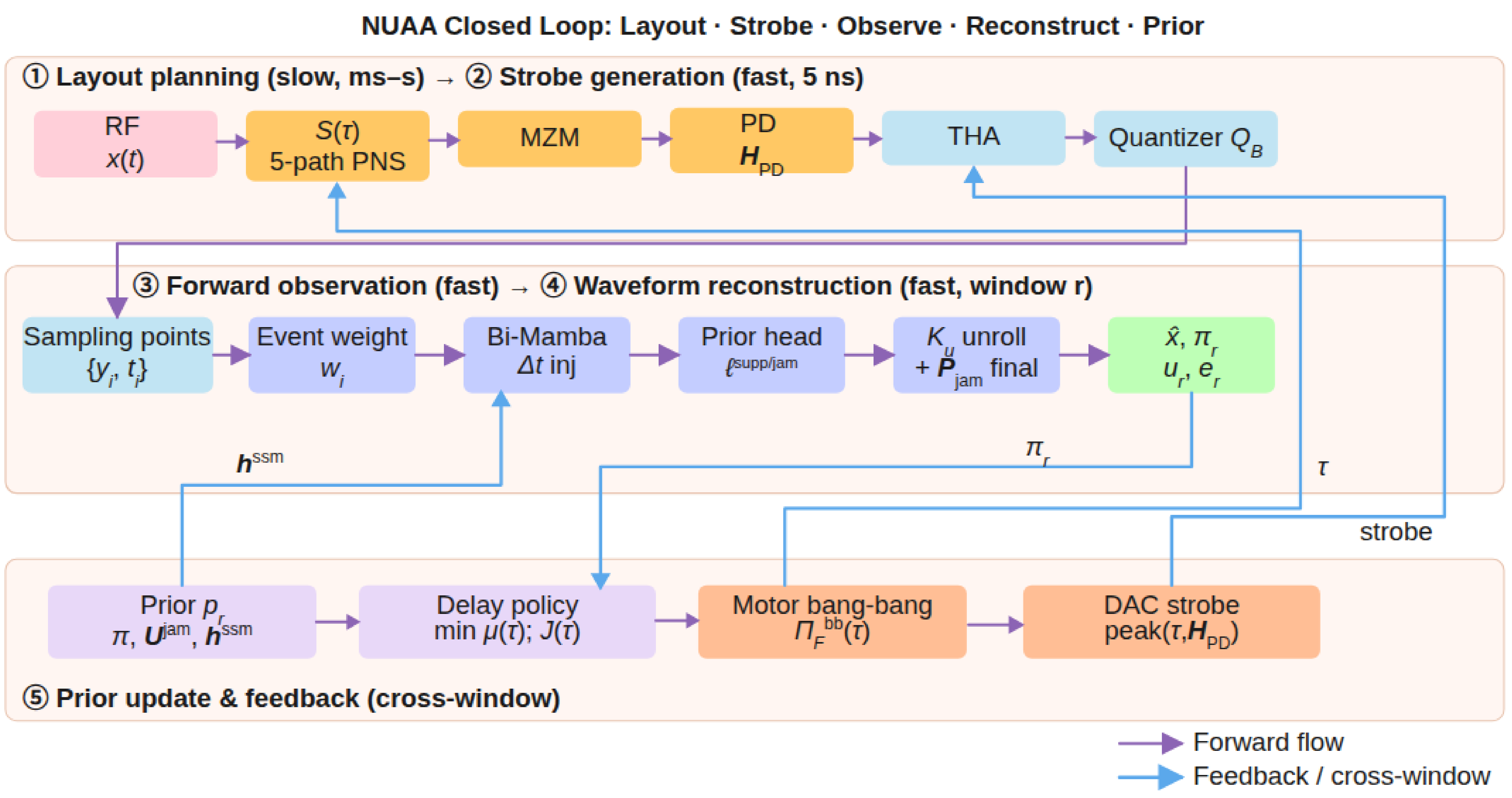

NUAA Closed Loop: Layout · Strobe · Observe · Reconstruct · Prior
① Layout planning (slow, ms–s) → ② Strobe generation (fast, 5 ns)
RF
$x(t)$
$S(\tau)$
5-path PNS
MZM
PD
$\boldsymbol{H}_{PD}$
THA
Quantizer Q_B
③ Forward observation (fast) → ④ Waveform reconstruction (fast, window r)
Sampling points
$\{y_i, t_i\}$
Event weight
w_i
Bi-Mamba
Δt inj
Prior head
$\ell^{supp/jam}$
K_u unroll
+ $\boldsymbol{P}_{jam}$ final
$\hat{x}, \pi_r$
u_r, e_r
$\boldsymbol{h}^{ssm}$
π_r
τ
strobe
Prior p_r
π, $\boldsymbol{U}^{jam}$, $\boldsymbol{h}^{ssm}$
Delay policy
min $\mu(\tau)$; $J(\tau)$
Motor bang-bang
$\Pi_F^{bb}(\tau)$
DAC strobe
peak(τ,$\boldsymbol{H}_{PD}$)
⑤ Prior update & feedback (cross-window)
Forward flow
Feedback / cross-window